Failure Analysis & Self Healing

R26-SE-039

Research Project Checklist

Ranasinghe S.I – IT22371768

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

Contents

INTRODUCTION 3

METHODOLOGY 4

SYSTEM ARCHITECTURE 5

EXPECTED OUTCOMES 5

REPO LINKS (GITHUB) 6

INTRODUCTION

Modern software development practices rely heavily on Continuous Integration and Continuous Deployment (CI/CD) pipelines to ensure fast and reliable software delivery. Automated testing frameworks such as Selenium, Cypress, and Playwright are widely used in these pipelines to detect software defects during development. Automated tests are executed continuously to verify that new code changes do not break existing system functionality. However, maintaining reliable automated test suites has become a significant challenge for software development teams. Automated tests frequently fail due to various reasons such as user interface changes, unstable element locators, synchronization issues, environment configuration problems, network instability, and inconsistent test data. These failures often require QA engineers to manually inspect logs, stack traces, and execution artifacts in order to identify the root cause of the problem. Studies have shown that a large percentage of automated test failures are not caused by real software defects but by unstable test scripts and environmental issues. This situation leads to increased debugging time, reduced efficiency of CI/CD pipelines, and higher maintenance costs for automated test suites.

To address these challenges, the NEXTGEN QA framework proposes an intelligent automated testing system consisting of four main components: Voice/NLP Intelligence, Test Generation and Execution Intelligence, Failure Analysis and Self-Healing Intelligence, and Quality Prediction with Requirement Traceability Matrix (RTM) generation. Within this framework, the Failure Analysis and Self-Healing component focuses on automatically analyzing test failures, identifying their root causes, and repairing unstable automated tests. Several existing automated testing tools such as Testim, Katalon Studio, and Functionize provide basic self-healing capabilities, particularly for repairing broken UI locators. However, most of these tools lack intelligent mechanisms for analyzing the root causes of failures or predicting unstable tests before they fail. Therefore, there is a need for a more advanced solution that combines machine learning-based failure analysis, predictive alerts, and automated test repair mechanisms without human intervention.

This research aligns with **Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure)** because it supports the development of more efficient and reliable software

infrastructure. It also aligns with the **Software Systems and Technologies research cluster**, which focuses on intelligent systems and software automation.

METHODOLOGY

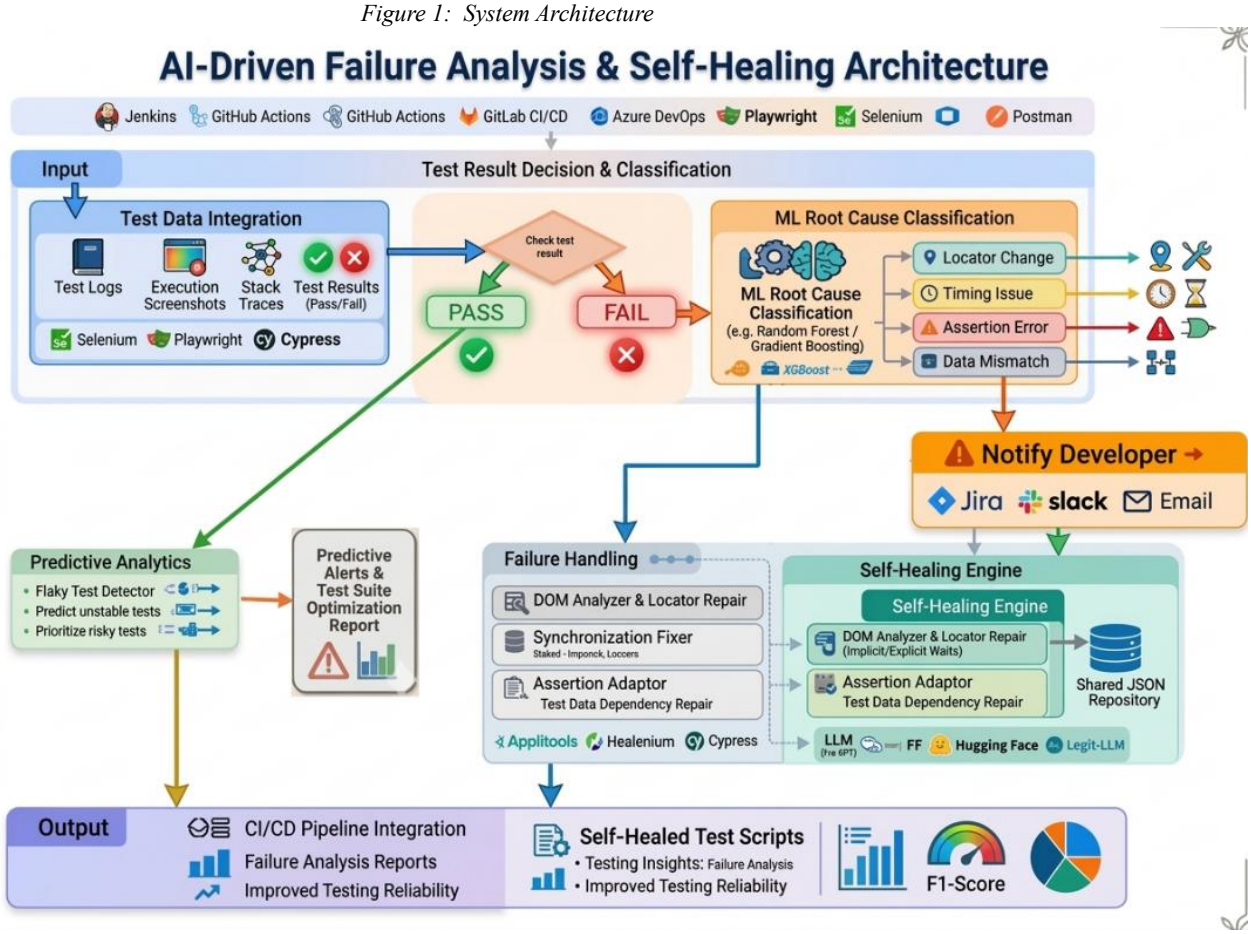
- This research adopts a **Design Science Research (DSR) approach** to design and evaluate an intelligent framework for failure analysis and self-healing in automated testing environments. Design Science Research is appropriate because the objective of this study is to create a technological artifact that solves a real-world problem faced in modern software development environments.

- The proposed solution addresses the instability of automated test suites used in Continuous Integration and Continuous Deployment (CI/CD) pipelines. The system will analyze test execution failures, classify their root causes using machine learning techniques, and automatically repair unstable test scripts. In addition, the system will detect flaky tests and improve test stability over time using adaptive learning mechanisms.

- The research process consists of the following stages:
  1. Collection of automated test execution artifacts
  2. Feature extraction and preprocessing
  3. Root cause classification using machine learning models
  4. Decision analysis and automated repair generation
  5. Predictive detection of flaky tests
  6. Adaptive improvement of automated test scripts
  7. Experimental evaluation of system performance

SYSTEM ARCHITECTURE

Figure 1: System Architecture



EXPECTED OUTCOMES

- **Root Cause Classification Accuracy**: Achieving high precision and F1-scores in classifying failure types using supervised ML models.
- **Automated Healing Success Rate**: Measuring the percentage of failed tests successfully repaired without human intervention.
- **Reduction in Debugging Time**: A measurable decrease in the manual effort required by QA engineers to inspect logs and fix scripts.
- **Flaky Test Identification**: Proactive detection of inconsistent test results using historical data trends.
- **Standardized Integration**: Modular CI/CD plugins for seamless deployment in enterprise DevOps environments.

REPO LINKS (GITHUB)

1. Frontend Application - https://github.com/R26-SE-039/frontend.git

2. Intelligent User Story Generator Service https://github.com/R26-SE-039/Failure_Analysis_Self_Healing.git